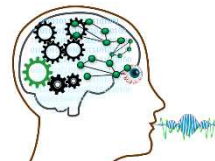




Vision A.I.

Software Engineering & Data Science
School of Engineering
University of St. Thomas

Dr. Chih Lai



SEIS 766: Syllabus

Vision A.I.

Instructor

Chih Lai, Ph.D.

E-mail: clai@stthomas.edu
(please include “**766**” in the subject field of your e-mail)

WWW: <http://www.linkedin.com/pub/chih-lai-ph-d/3/2b6/193>

Canvas: canvas.stthomas.edu

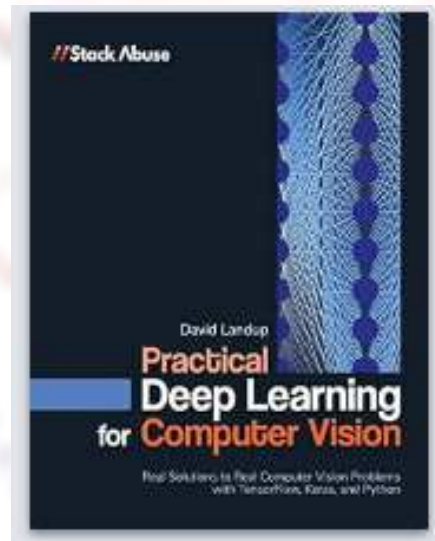
Office: OSS 308, Online Synchronous

Office Hours: 4:00 – 4:45 PM Thursday, Online, Zoom Links on Canvas

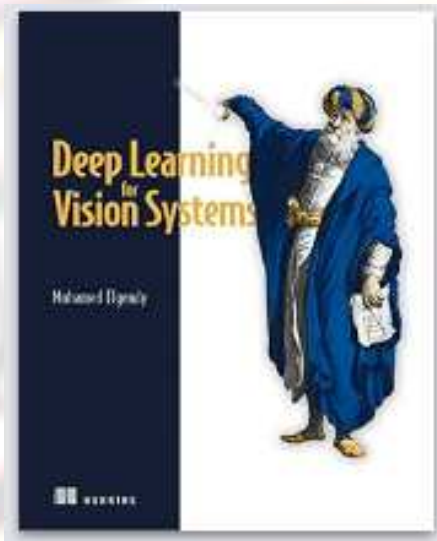
Class Rooms / Hours

5:30 – 8:30 PM Thursday, **SCC 314**
Zoom Links on Canvas

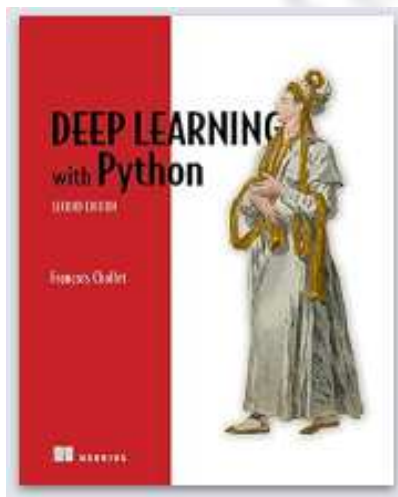
Textbooks



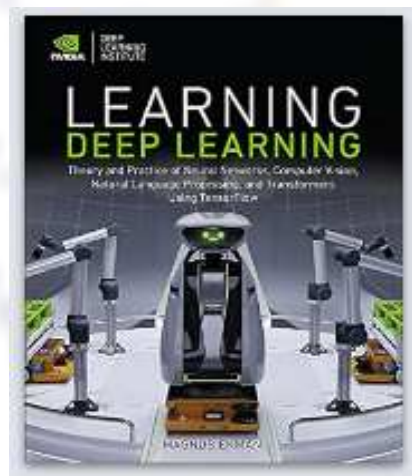
PDLCV



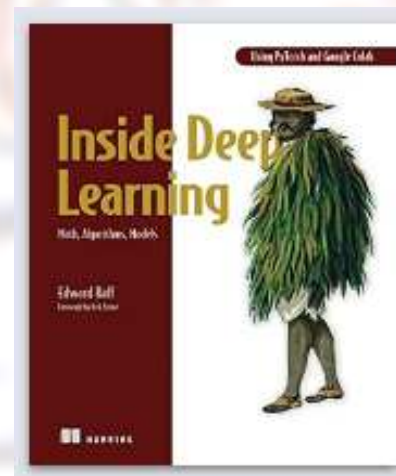
DLVS



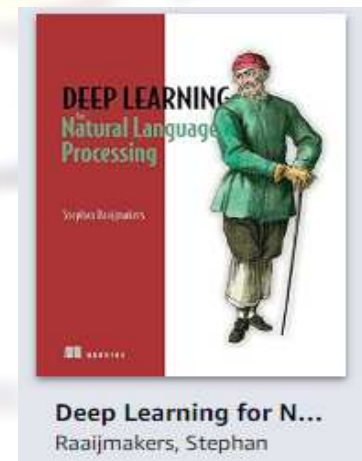
DLPy



LDL

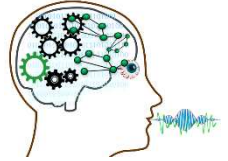


IDL



DLNLP

Course Description



- This course offers an interactive learning experience that delves into how machines perceive, analyze, and react to images and visual cues. You'll gain a greater understanding of images, videos, and their processing algorithms through hands-on exercises. By working on practical tasks like manipulating images and experimenting with Generative A.I. models like GANs, you'll discover the vast applications of Vision AI. Industries such as entertainment and healthcare are already benefiting from these technologies, which enable machines to recognize patterns, predict outcomes, and even create useful outputs. With this course, you'll learn both the theoretical and practical aspects of Vision AI, empowering you to combine your creativity with cutting-edge technology. At the end of this course, students will develop skill sets in visual intelligence and be poised to shape the future of this exciting field.

- **Prerequisite→** **SEIS 764 Artificial Intelligence**



- **Note→** This is ****NOT**** a high-level end-user tool-training class.
 - Google Tensor Flow (TF), Keras, Matlab. **NOT** training try-&-error, B-&-F programmers.
 - **Expectation→** 3-hour off-course study for every credit hour. (to meet expectation)
 - Many techniques were published in recent few years, including 2020 -- 2023.

Tentative Class Schedule– Session 1~4



No	Date	Topics
1	2 / 5 Intro.	* Vision AI, visual embedding (Z) / retrieval, object identification, image generation, dreaming / faking * Generative Adversarial Network (GAN), GAN variations, CycleGAN, UNIT GAN, CLIP * Auto-Encoder (AE), variation AE, image translation & apps, style transfer, quality evaluation * Visual AI on numeric business data???
2	2 / 12 Color Visual Z	* Image Processing Basic: MatLab, Python OpenCV, color / gray / black-white images, image process * Color space, RGB, HSV, YCbCr, indexed colors, color histogram & equalization, color motion map * Visual embedding Z for image retrieval / recommender, SMOTE on CNN Z-Code
3	2 / 19 Obj Detect	* NN / CNN with 1+ data sources, concatenation layer, augmenting multiple inputs * Object Detection, Semantic Segmentation, R-CNN, Masked RCNN, YOLO (You Look Only Once) * Conv / Deconv (Transpose), Pixel Classification, Pixel Labeling, Up-Pooling, CNN Encoder/Decoder * Augment 1+ sources, Case Study fish embryo, detect organs & morphology change, evaluation / IoU * Detect important parts of images, Occlusion Map, Pose estimation, POSE-net, face, fingerprint
4	2 / 26 AE	* Auto-Encoder/decoder (AE), Un/semi-supervised NN, unbalanced / few / NO label Y, * Usages: outlier detection, denoising, sparse AE, training VERY deep NN, fine tuning, * AE vs. PCA, creating fish embryo growing video, AE programming,

Tentative Class Schedule– Session 5~8



	Date	Topics
5	3 / 5 Train Loop VAE	* 2 input sources and concatenation, functional API, augmentation, data splitting, batches, shuffling, * Custom: training loop, loss function, update θs, shuffle, batches, data format on GPU * VAE basic idea
6	3 / 12 VAE	* Variational AE, encoder / decoder, 2 loss functions, KL loss, diverge, * VAE generating images AND numeric records / signals, Evaluating generation quality * Walk in latent Z space to adjust generate images
<u>7</u>	<u>3 / 19</u>	*Mid-term exam (5:30—7:30pm) *Project plan due Honorlock: https://services.stthomas.edu/TDClient/1898/ClientPortal/KB/ArticleDet?ID=152908
8	3 / 26 GAN CGAN WGAN	* Generative Adversarial Networks / GAN, generator G, discriminator D, GAN loss for G and D * Adversarial loss, MinMax loss, falsely label flipping, monitor training, implementation, mode collapse * Conditional GAN, <i>Wassertein</i> GAN (WGAN), WGAN regularization * GAN / CGAN generating images AND numeric records / signals, * Evaluating generation quality

Tentative Class Schedule– Session 9~10



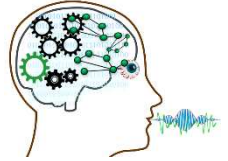
	Date	Topics
9	4 / 2 CLIP	* CLIP (Contrastive Language-Image Pre-training), * Multi-Modal Model, NO explicit training label, Multilabel Classification * Zero/one/many-shot prediction, Temperature, Contrastive Loss, experiments, GPU data format * Compare CLIP to BLIP-2 (Bootstrap-LIP), Image Captioning
10	4 / 9 Image Xslation	* Image-to-image translation across domains, CycleGAN, Translating / Imputing numeric records * CycleGAN loss functions: cycle consistency loss, adversarial loss, identity loss * UNIT GAN (UNsupervised Image-to-image Translation), 2 inputs and 4 outputs, architecture * Compare CycleGAN with UNIT, PatchGAN, pix2pixHD flow, applications for image translation
11	4 / 16 KD / SD SSL / BAN EMA	* Knowledge Distillation, Self-Distillation, Self-Supervised Learning * Teacher-Student Learning, Born-Again Network (BAN), * EMA / Exponential Moving Average Teachers (Meta DINO, DINO v2)
	Style Xfer Diffusion	* Neural Style Transfer (NST), content and style, loss, fast style transfer network * DeepDream, algorithm, pyramid levels / sizes / ratios, Octaves, loss/gradient ascent * Stable Diffusion, diffusion network, Swish activation

Tentative Class Schedule– Session 11~14



	Date	Topics
12	4 / 23 Quality	* Reference evaluation: L2, MSE, Structural Similarity (SSIM), Peak Signal-to-Noise Ratio (PSNR) * Non-Reference evaluation: Natural Image Quality Evaluator (NIQE), PIQE * Blind/Referenceless Image Spatial Quality Evaluator (BRISQUE), * Inception Score (IS), Frechet Inception Distance (FID), Wasserstein-2 Distance * <u>Perplexity</u>, binary cross entropy, categorical cross entropy, sparse cross entropy
	Image Process	* Image processing using AI, adjust brightness / darkness, histogram equalization, image de-noising * Image restoration, SISR(Single Image Super-Resolution), VDSR (Very-Deep Super-Resolution)
	Morphology	* Morphology: erosion, dilation, open, close, edge detection, 4/8-connectivity, low/high pass filter * Region properties, major axis, orientation, eccentricity, solidity, convex hull. Case study: retinal image
	Video Motion	* Video motion analysis (mouse movement case study), tracking objects, * Motions in Telemedicine Videos, Case Study: Knee Angle Prediction for Spinal Cord Injury Patients, * Virtual reality gaming, Siamese Network, Inception Network, 1×1 Conv,
13	4 / 30	*Project presentation (*Notes prepared by teams, <u>slides due before presentation</u>) Over Zoom
<u>14</u>	<u>5 / 7</u>	* Final exam * Comprehensive exam (*Final Project due 5 / 15)

Exams and Grading



Exams

The exams will be based primarily on the materials covered in class.

Grading

Homework + Participation	15%	(NO late submission after submission closed)
Project	30%	
Midterm exam	25%	
Final exam	30%	

Letter grade will be assigned **approximately** as follows:

80% — 100%	A, A-	(truly exceptional, exceed expectations)
70% — 80%	B+, B, B-	(meet expectations)
60% — 70%	C+, C, C-	
Below 60%	F	

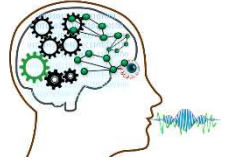


** Final distribution may be adjusted based on the class performance.

** Students who do **NOT** take exam(s) or miss project presentation will receive an **“F”** grade.

** Individual exceptions to the scheduled exams should be both **“individual and rare”**.

Course Project



Course Research Project

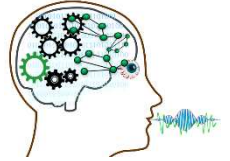
You will conduct an *Vision AI* project in a team of **5-6** people.

Check the class schedule for the due dates of project plan and final report.

■ Suggestions on Presentations

- Motivations– Give examples why the problem is interesting and important.
- Technical contents– Use examples to show how the techniques work.
- Discussion– Pros & cons, pitfalls.
- Performance studies (precision and speed)
- References– Background and related work
- Issues/impacts related to information ethics and privacy

Project Plan and Final Project Submission

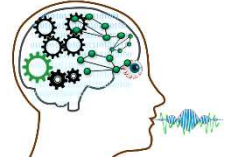


- Each team must submit a final zipped file w/ the following sections:

1. **Project Title**
2. **Description of data source and web link(s)**
3. **# of records & # of attributes with description of each attribute**
4. **Some general statistics of the dataset with few sample records (or images or text)**
5. **Tools / methods / computing platform(s) your team (plan to) use in your study**
6. **Detail description of what problems/questions your team plans to predict / study**
7. **Final presentation slides** (methods used, comparisons, conclusions) **← w/ page number**
8. **Project processed datasets (or link)**
9. **Project code / programs with a (short) user manual**

- Zip item 1-9 into a ****single**** zipped file w/ **Project Title** as the name.

- **ONE** team member shares the zipped file on **UST OneDrive** with the instructor.
- Please verify that the shared link is accessible by all your team members (hence the instructor).
- DO ****NOT**** send your **BIG** zipped file via email to the instructor.



Checklist Before Presentation

- If your team plans to use the classroom PC for your presentation, please **login to that PC at least few hours before** your team's presentation.
- Otherwise, it will take a while to login for the first time before your presentation, and it will delay other team's presentation.
- If your team plans to use your own laptop for your presentation, please test the connection between your laptop and the classroom projector **at least few weeks before** your team's presentation.
- You may need to find an appropriate connector to connect to the projector.
- Please **carefully** control your team's presentation time.
 - Consider possible questions that may interrupt your presentation.
 - Please do **NOT "READ"** your slides.
 - **Please add page numbers to your slides**

Resources



Computing Resources

OSS 327 Computer Lab,

Please check your UST e-mail account regularly.

Google Colab (Python)

<https://research.google.com/colaboratory/>

Matlab Online

https://matlab.mathworks.com/?s_tid=tah_po_start

Matlab Local

<https://www.mathworks.com/academia/tah-portal/university-of-st-thomas-31483492.html>

https://www.mathworks.com/store/link/products/student/new?s_iid=htb_buy_gtwy_cta3

Support Staff

School of Engineering Tech Group (soetech@groups.stthomas.edu) for questions on using the equipment in Room 327.

Attendance Policy

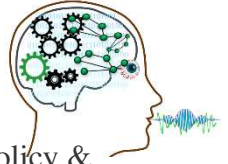
Course attendance is expected, but no grade is given for it. Students who miss sessions are responsible for all information in that session. Students who need to miss **presentations** or **exams** due to unavoidable conflicts must arrange in advance to make up the session with the instructor.

Course Assignments

Homework will be assigned from time to time during the semester in order to reinforce the concepts/techniques discussed in the class. Assignments will be collected on the specified due dates. **NO** late submission will be accepted without proper reasons.

ENHANCEMENT PROGRAM FOR DISABILITY: In compliance w/ the University of St. Thomas policy & disability guidelines, your instructors are available to discuss appropriate academic accommodations that you may require as a student with a disability. Requests for academic accommodation should be made during the first week of the semester so timely arrangements can be made. Students are encouraged to register with the Enhancement Program for disability verification and for determination of reasonable accommodations. Classroom accommodations will be provided to students with documented disabilities. Info about the Enhancement Program is available at <http://www.stthomas.edu/enhancementprog/> or (651) 962-6315.

Policy Statement



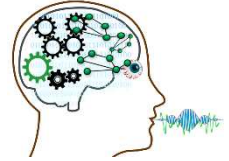
- **ENHANCEMENT PROGRAM FOR DISABILITY:** In compliance w/ the University of St. Thomas policy & disability guidelines, your instructors are available to discuss appropriate academic accommodations that you may require as a student with a disability. Requests for academic accommodation should be made during the first week of the semester so timely arrangements can be made. Students are encouraged to register with the Enhancement Program for disability verification and for determination of reasonable accommodations. Classroom accommodations will be provided to students with documented disabilities. Info about the Enhancement Program is available at <http://www.stthomas.edu/enhancementprog/> or (651) 962-6315.

- **Academic Dishonesty Policy:** Honesty and trust among students and between students and faculty are essential for a strong, functioning academic community. Students are expected to complete all academic work on their own. Academic dishonesty is not tolerated in the Department of Software Engineering & Data Science at the University of St. Thomas.
Students are required to complete/submit all academic work as one's own. If a student fails to do so, a grade of zero is issued. If the student commits a second academic dishonesty violation in the same or future course, a course grade of F is issued. If a third violation occurs, the student is dismissed from the program.

Common forms of academic dishonesty include, but are not limited to, the following:
 - Cheating: Using, accessing, or trying to use unauthorized materials, information, or study aids in any academic work submitted for credit; giving or receiving unauthorized materials or aid in an academic exercise.
 - Fabricating or misrepresenting: Falsifying or inventing any information or citation in an academic exercise; fabricating or misrepresenting a reason for an absence or missed deadline; submitting academic work completed largely in one class to satisfy an assignment in a second class without instructor permission; fabricating or tampering with academic records.
 - Facilitating Academic Dishonesty: Knowingly helping or attempting to help another student commit academic dishonesty.
 - Plagiarizing: Claiming or submitting the ideas, words, or statements of another person as one's own without proper citation regardless of intent to deceive. Even inadvertent plagiarism is a serious matter; students must familiarize themselves with citation standards.

- **Exam Proctoring Policy:** The University of St. Thomas continues to strive for academic excellence and integrity in times of continuous technological advancement. The University has selected third-party online exam monitoring tools (“tools”) to assist in combating academic dishonesty in the online testing environment. This is part of a three-step approach of prevention, detection, and communication. By registering for a course within the School of Engineering’s Graduate Programs in Engineering and Software, students are agreeing to install the tools on exam-taking devices for the duration of any course exam, if required by the faculty instructor.

- **DISCLAIMER:** This syllabus is not a contract, but a plan for action. The instructors reserve the right to alter its stipulations, upon prior notification to students, if and when educational circumstances warrant changes.



■ **Academic Experience:**

Dr. Lai is a professor with GPS. He has taught courses in Artificial Intelligence, Machine Learning & Machine Vision, Deep Learning, Healthcare Analytics. Dr. Lai is also a exchange / visiting scientist at the Helmholtz National Research Center in Germany in 2019. He was invited as a visiting professor of the Informatics Department at Trier University of Applied Science in Germany in 2010.

■ **Research and Publications:**

Dr. Lai's research interests include Artificial Intelligence, Machine Learning and Deep Learning on multimedia data (numerical, images, videos, text). Dr. Lai has published many technical papers in IEEE and ACM conferences / journals. Dr. Lai received grants over \$1.1-million from National Science Funding (NSF), State of Minnesota, Amazon, Microsoft, and German government. Please visit <http://www.linkedin.com/pub/chih-lai-ph-d/3/2b6/193> for details.

■ **Industry Experience:**

Before joining UST, Dr. Lai was a principal software engineer, working on a new aircraft collision avoidance system (ADS-B) which FAA has mandatory installation on most aircraft by 2020. From this project, Dr. Lai received three U.S. patents and three European patents. Dr. Lai also worked with Medtronic and has patents on monitoring & evaluating Parkinson patients after deep-brain stimulation surgery. Dr. Lai is working with UoM on Surgical Images to guide surgical procedure (patent app). Other industry experience includes applying A.I. in counterfeit product detection on smartphone & cloud (patented), predicting power generation and equipment failures for hydro/wind energy industry.